Data cleaning process and organization of data with SQL

Session 4

Agenda

- **Verifying Data Cleaning Results**

Introduction to Verifying Data Cleaning Results

Verifying and validating data after cleaning is crucial to ensure accuracy, completeness, and reliability. This process involves:

- **Techniques for Validation:** Methods to check that data cleaning tasks have been properly executed.
- **SQL Queries for Comparison:** Queries to compare data before and after cleaning.
- **Quality Assurance:** Strategies to audit SQL results and verify data integrity.

These practices help in confirming that the data cleaning process has met the intended objectives and that the resulting data is ready for analysis.

Techniques for Verifying Data Cleaning Results

1- **Data Profiling:** Analyze the data to understand its structure, content, and quality before and after cleaning.

Example: Generate summary statistics, and frequency distributions, and identify anomalies.

2- **Comparison with Original Data:** Compare cleaned data with the original dataset to ensure that no critical information was lost and that cleaning tasks were applied correctly.

Example: Check row counts, key metrics, and specific data points.

Techniques for Verifying Data Cleaning Results

3- Validation Rules: Apply business rules and logic to validate cleaned data.

Example: Verify that age values fall within a reasonable range or that email addresses follow the correct format.

4- **Consistency Checks:** Ensure that data is consistent across different tables and datasets.

Example: Cross-verify foreign key relationships and data aggregations.

SQL Queries to Compare Data Before and After Cleaning

1. **Comparing Row Counts:**

```
SELECT COUNT(*) AS row_count FROM original_data;
SELECT COUNT(*) AS row_count FROM cleaned_data;
```

SQL Queries to Compare Data Before and After Cleaning

2. Checking for Null Values:

```
SELECT COUNT(*) AS null_count
FROM original_data
WHERE column_name IS NULL;
```

```
SELECT COUNT(*) AS null_count
FROM cleaned_data
WHERE column_name IS NULL;
```

SQL Queries to Compare Data Before and After Cleaning

3. Verifying Data Integrity

```sql
SELECT * FROM original_data
WHERE column_name IS NULL OR column_name = '';
```

```sql
SELECT * FROM cleaned_data
WHERE column_name IS NULL OR column_name = '';
```

SQL Queries to Compare Data Before and After Cleaning

4. Identifying Duplicate Records:

```sql
SELECT column_name, COUNT(*)
FROM original_data
GROUP BY column_name
HAVING COUNT(*) > 1;
```

Common Pitfalls and Errors

1- **Incomplete Cleaning:** Ensure that all intended cleaning operations were applied.

Error Example: Missing values or inconsistent formatting due to incomplete operations.

2- **Data Loss:** Verify that no important data was accidentally removed or altered.

Error Example: Data truncation or unintended deletions.

Common Pitfalls and Errors

3- **Incorrect Validation Rules:** Ensure that validation rules reflect actual business requirements.

Error Example: Applying too restrictive rules that exclude valid data.

4- **Cross-Table Consistency:** Confirm that data relationships and integrity are maintained across tables.

Error Example: Mismatched foreign key references or missing related records.

Practical Session - End-to-End Data Cleaning Process

1- **Identify Data Issues:**

- Dataset: ecommerce_transactions
- Issues to Identify: Missing values, duplicates, incorrect formats.

2- **Perform Data Cleaning:**

- Tasks: Apply cleaning operations using SQL functions like TRIM, REPLACE, and COALESCE.

3- **Validate and Compare Results:**

- Compare Data: Use queries to check the results before and after cleaning.
- Validation Rules: Ensure that the cleaned data meets specified rules and criteria.

Practical Session - End-to-End Data Cleaning Process

1- Load Dataset:

--Load the dataset into a SQL environment.

```
SELECT * FROM ecommerce_transactions;
```

2- Clean Data:

- Remove leading/trailing spaces, replace null values.

```
UPDATE ecommerce_transactions
SET transaction_amount = COALESCE(TRIM(transaction_amount), '0');
```

3- Validate Data:

```
-- Check for remaining issues.
SELECT COUNT(*) AS null_count
FROM ecommerce_transactions
WHERE transaction_amount IS NULL;
-- Compare row counts.
SELECT COUNT(*) AS row_count FROM original_ecommerce_transactions;
SELECT COUNT(*) AS row_count FROM cleaned_ecommerce_transactions;
```

4- Generate Reports: Create summary reports to show the impact of cleaning operations.

```sql
-- Summary of transactions before cleaning
SELECT transaction_category, COUNT(*) AS original_count
FROM original_ecommerce_transactions
GROUP BY transaction_category;

-- Summary of transactions after cleaning
SELECT transaction_category, COUNT(*) AS cleaned_count
FROM cleaned_ecommerce_transactions
GROUP BY transaction_category;
```

Any Questions ?

Thank You